

Lab 02 - First Verilog

In this lab, you've learned how to do an initial and simple design in Verilog to learn the Vivado tooling and process involved in RTL/FPGA design.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab Summary

Summarize your learnings from the lab here.

Make sure you run a simulation every time you make a change. If you mess up the type of the sources you can change them in the project manager. Make sure the board is turned on. Dont worry if the number of projects is off since it is based on the cores of your cpu.

Lab Questions

1 - Describe the stages of building a Verilog project in Vivado.

You create a project in Vivado, select where you want the project to be. Then add source and/or constraint files. After that choose the board you are going to use. Run a behavior simulation. Check the simulation to make sure the input is correct. Plug the board into the computer. Generate the bitstream (make sure the board is turned on). Open the hardware manager, then connect the board. After that run the program to the device.

2 - What is the value in looking at the elaborated design schematic?

Allows you to see all the switches and leds. Which will allow you to trouble shoot if anything is wrong.

3 - Why should we simulate our designs frequently? What does the simulation do?

Allows us to find any errors. Also allows us to see what is working. Also allows the program to accept the changes that were made. As well as allows for the program to open up a new instance of the simulation.

Code Submission

Upload a .zip of all your code or a public repository on GitHub.

https://github.com/JaredLackerman/ECE230L-Lab-2/blob/main/week_2_first_verilog.zip